# An AI-Enabled Intelligent Study Assistant to Enhance Student Learning Outcomes

Emotion and Progress-Aware Adaptive Study Reminder System

R26-SE-022

## Project Proposal Report

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## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where acknowledgment is made in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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# ABSTRACT

Using a Design Science Research (DSR) approach, the system integrates three core technical modules: an extended SM-2 spaced repetition model that generates per-item retention priority queues; an Emotional State Fusion Module that computes a composite Readiness Score (High / Medium / Low) from embeddings supplied by a sibling emotion detection component; and a Reinforcement Learning (RL) scheduling agent that selects the optimal action — send, delay, or skip — refined continuously through student accept, snooze, and dismiss feedback. A 4-week controlled study with 30 SLIIT undergraduates will validate results against targets of >85% adherence improvement, $p<0.05$ retention gain, reduced NASA-TLX cognitive fatigue, and SUS score >70. This work is the first to embed affective readiness directly into adaptive reminder scheduling, offering a human-centred alternative to purely algorithmic study tools with broad applicability across university education, online learning platforms, and corporate training.

University students increasingly struggle with academic stress, burnout, and poor knowledge retention because existing study reminder and scheduling tools are entirely blind to their emotional and cognitive states. Current spaced repetition systems such as Anki and Quizlet schedule sessions purely on performance metrics — quiz scores and completion rates — with no awareness of a learner's emotional readiness, mental fatigue, or motivational state. This research addresses that gap by designing and implementing an Emotion and Progress Aware Adaptive Study Reminder System, an individual component of the broader Smart Study Assistant platform (R26-SE-022), which intelligently schedules personalized study reminders by fusing emotional state signals, memory decay models, and behavioral interaction history to maximize study adherence and knowledge retention while minimizing cognitive fatigue.

*Keywords: Adaptive Study Reminders, Emotion-Aware Scheduling, Spaced Repetition, Memory Decay Models, Affective Computing, Reinforcement Learning, Student Wellbeing, Personalized Learning.*

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# LIST OF ABBREVIATIONS

| Abbreviation | Full Form |
|---|---|
| DSR | Design Science Research |
| Web Push API | Web Notification API (browser push) |
| LLM | Large Language Model |
| ML | Machine Learning |
| NASA-TLX | NASA Task Load Index |
| RL | Reinforcement Learning |
| SLIIT | Sri Lanka Institute of Information Technology |
| SM-2 | SuperMemo Algorithm 2 |
| SUS | System Usability Scale |
| TAF | Topic Assessment Form |

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# 1. INTRODUCTION

## 1.1 Background and Context

The academic environment for university students has grown increasingly demanding, with intensified workloads, competitive examinations, and heightened expectations contributing to widespread stress and burnout [1]. Research consistently demonstrates that a student's emotional and cognitive state at the time of study directly influences learning effectiveness, knowledge retention, and academic performance. Despite this well-established relationship, the vast majority of existing study management tools and reminder systems are designed without any awareness of the learner's emotional state.

Current study planning platforms, such as spaced repetition systems like Anki or Quizlet, schedule review sessions solely based on performance metrics — how well a student answered a question previously. While these systems have proven effective for rote memorization, they fundamentally fail to account for learner-specific factors such as emotional readiness, mental fatigue, or motivational state [2]. A student who is anxious, frustrated, or cognitively overloaded is unlikely to benefit from a reminder that arrives at the algorithmically optimal interval; indeed, such reminders may increase procrastination and disengagement [3].

This research is situated within the broader Smart Study Assistant project (R26-SE-022), which combines an LLM-based tutoring module, a learning-objective-aware summarization system, a privacy-preserving emotion detection module, and this component — the Emotion and Progress Aware Adaptive Reminder System. The societal importance of this research aligns with the United Nations Sustainable Development Goal 4 (Quality Education).

## 1.2 Literature Review

Research on adaptive learning and intelligent tutoring systems has advanced significantly over the past decade. Mehta and Sharma [2] explored AI integration through PrepPal, demonstrating improvements in engagement through AI-driven content delivery, but without affective signals. Liu et al. [4] developed LPITutor using RAG and prompt engineering, improving factual accuracy but lacking real-time emotional adaptation. Spaced repetition, formalized through the Ebbinghaus forgetting curve and the SuperMemo SM-2 algorithm [5], operates on the assumption of cognitive availability — assuming the student is always ready to receive information at the scheduled time.

Picard et al. [6] established foundational principles for affective computing, demonstrating that emotional signals are actionable inputs for adaptive systems. Kar et al. [1] showed in the MathBuddy system that incorporating facial expression analysis into tutoring improves pedagogical effectiveness. Fetaji et al. [3] further validated that systems adapting to learner cognitive states outperform static counterparts in both retention and satisfaction. Sweller's cognitive load theory [7] emphasizes that exceeding working memory capacity leads to cognitive overload and reduced learning.

## 1.3 Research Gap

A systematic review of existing adaptive learning and reminder systems reveals a significant and unaddressed research gap: no published study has integrated emotional readiness signals directly into adaptive study reminder scheduling. Spaced repetition models such as SM-2 [5] and knowledge-tracing approaches [9, 11] optimise review timing solely from performance history, disregarding the affective dimension of learning. Affective computing research [6] confirms that emotional state is a strong predictor of learning effectiveness, yet this insight remains unexploited in reminder delivery systems. Reinforcement learning has been explored for educational recommendation [9] but not for emotion-conditioned reminder scheduling. The research scope of this component is therefore bounded to the scheduling and notification layer of the Smart Study Assistant — specifically, determining when, how often, and in what content modality to deliver a study reminder given a student's current emotional and cognitive state. This sub-problem has not been investigated in prior published work, establishing clear academic novelty and contribution.

### 1.3.1 Research Value

The table below summarises the primary research gaps across four key dimensions, forming the basis for both the research and product contributions of this work.

| Dimension | Existing Approaches | Identified Gap |
|---|---|---|
| Scheduling Basis | Performance metrics only (quiz scores, completion) | Ignores emotional readiness and cognitive fatigue |
| Affective Integration | Emotion detection exists in isolation | No integration into scheduling logic |
| Reminder Content | Static content type per session | No dynamic content adaptation based on affect |
| Adaptiveness | Rule-based or static ML models | No real-time RL-based adaptive decision-making |

*Table 1. Research Gap Summary in Adaptive Study Reminder Systems*

### 1.3.2 Product Value

From a commercial and product standpoint, no existing EdTech tool combines emotion-aware scheduling with memory decay modelling in a single, deployable web-based reminder system. Tools such as Anki, Quizlet, and Notion Study Plans deliver reminders based on fixed schedules or performance metrics alone; none incorporate affective readiness signals or adapt content type to the learner's current cognitive-emotional state. Forest and Focusmate address student productivity but offer no subject-specific scheduling or emotional intelligence. This represents a clear and exploitable product gap.

The product scope of this component targets undergraduate university students in Sri Lanka and the broader South Asian region, where adoption of AI-based learning tools is growing rapidly but wellbeing-centered features remain entirely absent from available products. The system is architected as an independently deployable microservice within the R26-SE-022 platform, exposing a documented REST API that enables future integration with third-party LMS platforms such as Moodle and Blackboard, as well as institutional student portals. This architecture ensures the reminder system can be licensed or white-labelled to universities without requiring full adoption of the parent platform.

Three distinct product advantages position this system for adoption. First, it is the only reminder system that conditions scheduling decisions on emotional readiness rather than performance history alone, directly reducing the problem of reminders arriving during high-stress or cognitively depleted states. Second, the Readiness Score computation is privacy-preserving by design - raw emotional embedding never persisted, which removes a major institutional adoption barrier around student data sensitivity. Third, the reinforcement learning agent continuously improves personalization over time from each student's own interaction history, meaning the product becomes more valuable the longer it is used, creating natural retention and switching costs for commercial deployment.

# 2. OBJECTIVES

## 2.1 Main Objective

To design and implement an Emotion and Progress Aware Adaptive Study Reminder System that integrates student emotional state signals with learning progress data and memory decay models to intelligently schedule personalized study reminders, thereby maximizing study adherence and knowledge retention while minimizing cognitive fatigue.

## 2.2 Specific Objectives

1. Model study progress and memory decay using an extended spaced repetition algorithm with item-difficulty parameters to generate dynamic retention probability estimates per content item.
2. Integrate emotional state embeddings from the emotion detection module into a composite Readiness Score through a multi-input fusion module.
3. Implement an Adaptive Scheduling Agent using reinforcement learning (contextual bandit) to optimize reminder timing, frequency, and content type.
4. Design a non-intrusive Reminder Delivery Interface supporting browser browser push notifications and in-page alert prompts, with user feedback collection.
5. Evaluate adherence, knowledge retention, and cognitive fatigue against static scheduling baselines through a 4-week controlled user study with 30 SLIIT students.

# 3. METHODOLOGY

## 3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, as the primary goal is to design, build, and evaluate an artefact — the Emotion and Progress Aware Adaptive Study Reminder System. DSR is well-suited because it combines rigorous design with practical evaluation, ensuring the system meets both technical performance and user experience requirements. The research follows an iterative cycle of design, prototype, and evaluation.

## 3.2 Technical Methods and Algorithms

1. Memory Decay and Progress Modeling

- An extended SM-2 algorithm models per-item retention probability as a function of review history, difficulty, and time elapsed.
- Item difficulty parameters are updated dynamically from student performance data.
- Retention probability estimates feed into the scheduling agent as input features.

2. Emotional State Fusion Module

- Receives normalized emotional embeddings from the sibling emotion detection module.
- Computes a composite Readiness Score using a weighted combination of valence, arousal, and attention indicators.
- Readiness Score is classified into High, Medium, or Low tiers to guide scheduling decisions.

3. Adaptive Scheduling Agent (Reinforcement Learning)

- Implements a contextual bandit / RL agent that takes a state vector (Readiness Score, item priorities, time of day, recent interaction history) and outputs an action: send reminder, delay, or skip session.
- Reward signal derived from student feedback: accept (+1), snooze (0), dismiss (-1).
- Attention-based sequence models capture temporal dependencies in study behavior.

4. Reminder Content Personalization

- Selects activity type based on cognitive state: Active Recall (high readiness), Guided Review (medium readiness), Passive Reading (low readiness).
- Maps content items to difficulty tiers and recommends appropriate review modality.

## 3.3 Tools and Platforms

| Tool / Platform | Purpose | Justification |
|---|---|---|
| Python + FastAPI | Backend & ML pipeline | Lightweight, scalable, widely used in AI systems |
| TensorFlow / PyTorch | RL agent & sequence models | Industry-standard deep learning frameworks |
| scikit-learn | Memory decay modeling | Mature ML library with SM-2 extension support |
| Spring Boot (Java) | RESTful API orchestration | Robust, enterprise-grade microservice framework |
| React.js | Cross-browser web interface | Single codebase for cross-browser web reminder delivery |
| MongoDB | Student profile & interaction logs | Flexible schema for behavioral event data |
| MySQL | Scheduling data store | Reliable relational store for structured scheduling records |
| Web Notification API (browser push) | Push notification delivery | Low-latency, reliable cross-platform notifications |
| AWS / Firebase Hosting | Cloud deployment | Scalable infrastructure with global availability |
| Docker / Kubernetes | Containerisation & orchestration | Consistent deployment and horizontal scalability |
| GitHub | Version control | Collaboration and reproducibility across team |

*Table 2. Tools and Platforms with Justification*

## 3.4 Data Collection and Ethical Considerations

Data Collection

- Study interaction logs (anonymized): reminder response events, session timestamps, quiz scores.
- Emotional embeddings received from the sibling emotion detection module (no raw facial/audio data stored).
- Self-reported mood surveys administered at the start and end of each user study week.

Ethical Considerations

- All participants provide written informed consent prior to enrollment in the user study.
- No raw biometric data is stored by this component; only aggregated Readiness Scores are persisted.
- All data is anonymized, encrypted at rest and in transit, and stored securely within the system boundary.
- The system complies with SLIIT's ethical research guidelines and responsible AI standards.
- Participants may withdraw from the study at any time without penalty.

## 3.5 Validation Strategy and Performance Metrics

Validation Strategy

- Technical Evaluation: Compare adherence rates between RL-based adaptive scheduling and static SM-2 baseline.
- A/B Testing: Emotion-aware scheduling vs. performance-only scheduling to quantify the impact of affective signals.
- User Evaluation: Controlled user study with 30 SLIIT students over 4 weeks, collecting NASA-TLX and SUS scores.

| Metric | Target | Measurement Method |
|---|---|---|
| Adherence Rate | > 85% vs. baseline | Reminder accept rate over 4-week study |
| Retention Improvement | $p < 0.05$ | Pre/post quiz score comparison |
| Cognitive Fatigue (NASA-TLX) | Lower than baseline | Weekly NASA-TLX surveys |
| False Positive Reminder Rate | < 15% | Proportion of reminders dismissed immediately |
| System Usability Score (SUS) | > 70 | Post-study SUS questionnaire |
| Readiness Score Accuracy | > 80% alignment | Comparison with self-reported mood survey |

*Table 3. Performance Metrics for Evaluation*

# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

## 4.1 Overview of the Proposed Solution

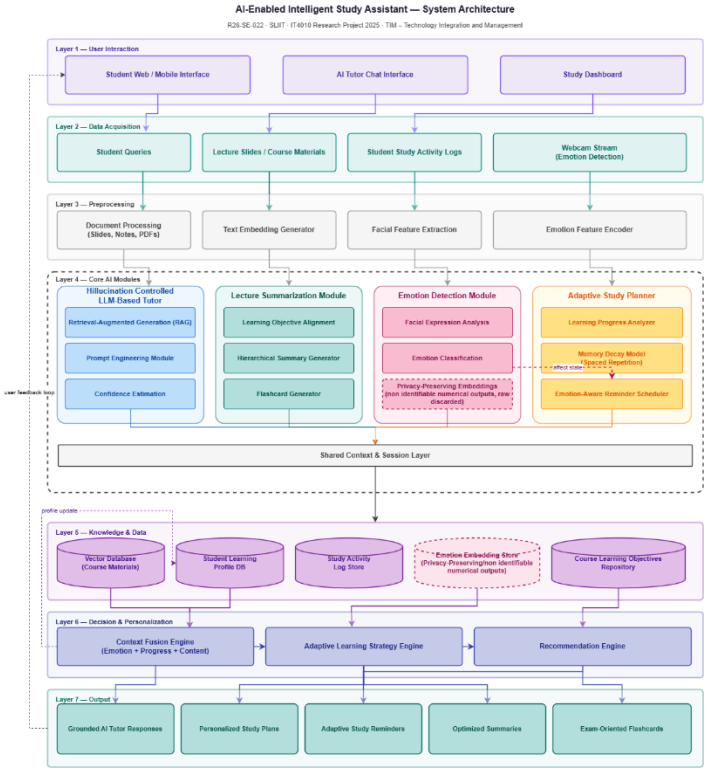



*Figure 1. Overall System Architecture – Smart Study Assistant (R26-SE-022)*

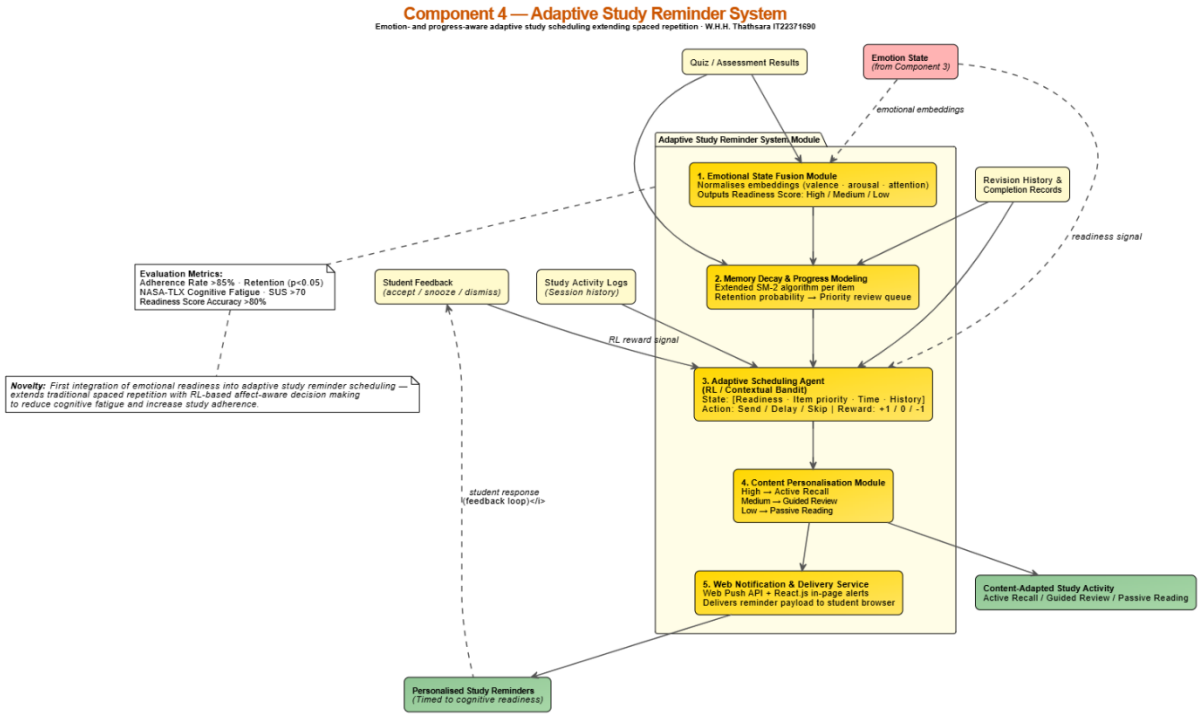



*Figure 2. Individual Component Architecture – Adaptive Study Reminder System*

## 4.2 System Components

1. Emotional State Fusion Module

- Receives normalized emotional embeddings from the sibling Emotion Detection Module via the Shared Context Layer.
- Computes a composite Readiness Score using valence, arousal, and attention indicators.
- Classifies the student's cognitive-emotional readiness into High, Medium, or Low tiers.

2. Memory Decay and Progress Modeling Module

- Maintains a per-student, per-item retention probability model using an extended SM-2 algorithm.
- Generates an ordered priority queue of content items due for review based on predicted forgetting.
- Updates item difficulty parameters dynamically from performance history.

3. Adaptive Scheduling Agent (RL / Contextual Bandit)

- Takes a composite state vector (Readiness Score, item priorities, time of day, interaction history) as input.
- Outputs an optimal action: send reminder now, delay by a specified interval, or skip the session.
- Uses student feedback (accept/snooze/dismiss) as the reward signal for online learning.

4. Reminder Content Personalization Module

- Selects appropriate study activity type based on the current Readiness Score tier.
- Maps item priority to the most effective review modality (active recall, guided review, or passive reading).

5. Notification and Delivery Service

- Delivers reminders via Web Notification API (browser push) (browser push notifications) and in-app prompts through the React web interface.
- Collects user feedback responses and forwards them to the scheduling agent as reward signals.

## 4.3 Component Interactions

The interaction flow between components follows a sequential pipeline:

1. The Emotion Detection Module pushes updated emotional embeddings to the Shared Context Layer.
2. The Emotional State Fusion Module retrieves these embeddings and computes the Readiness Score.
3. The Memory Decay Module reads the student's study history and generates an item priority queue.
4. The Adaptive Scheduling Agent receives both the Readiness Score and item priorities, then selects the optimal action.
5. The Content Personalization Module selects the activity type and prepares the reminder payload.
6. The Notification Service delivers the reminder to the student's browser.
7. The student responds (accept/snooze/dismiss), and the feedback is returned as a reward to update the RL agent.

## 4.4 Data Flow Explanation

Input Flow:

- Emotional embeddings from Emotion Detection Module → Fusion Module → Readiness Score
- Study history and quiz performance → Memory Decay Module → Item priority queue
- Student feedback (accept/snooze/dismiss) → RL agent reward signal

Processing Flow:

- Readiness Score + Item priorities + Context features → State vector for RL agent
- RL agent applies learned policy → Action selection (send / delay / skip)
- Selected action + Readiness tier → Content type selection → Reminder payload

Output Flow:

- Reminder payload → Web Notification API (browser push) → Student web browser
- Student response → Feedback loop → RL agent model update

The data flow ensures the system always prioritizes the student's cognitive readiness over rigid algorithmic schedules, delivering timely and contextually appropriate reminders.

## 4.5 Scalability and Security Considerations

Scalability

- The microservice architecture allows horizontal scaling of individual components independently.
- MongoDB and MySQL databases support sharding and read replicas for high-concurrency scenarios.
- Containerization via Docker and Kubernetes ensures elastic resource allocation.
- The system is designed to support at least 500 concurrent student users with sub-500ms scheduling latency.

Security

- All API communication is secured using JWT-based authentication and HTTPS encryption.
- Role-Based Access Control (RBAC) restricts data access to authorized services only.
- Emotional embeddings are not stored in raw form; only aggregated Readiness Scores are persisted.
- All personally identifiable information is encrypted at rest (AES-256) and in transit (TLS 1.3).

# 5. USER REQUIREMENTS

## 5.1 Stakeholder Identification

| Stakeholder | Role | Interest / Responsibility |
|---|---|---|
| University Students | Primary users | Use the reminder system to manage study schedules and receive adaptive reminders based on emotional state |
| Lecturers / Academic Staff | Content providers | Provide course materials used for content recommendation and scheduling |
| Research Team / Developers | Designers & implementers | Build, test, and maintain the adaptive scheduling system |
| Supervisors / Evaluators | Quality assurance | Ensure the system meets academic, ethical, and technical standards |
| SLIIT IT Department | Deployment support | Provide infrastructure support for hosting, integration, and security compliance |

*Table 4. Stakeholder Identification*

## 5.2 Functional Requirements

| ID | Requirement | Description |
|---|---|---|
| FR1 | Memory Decay Modeling | Model per-item memory decay and generate retention probability estimates using extended SM-2. |
| FR2 | Emotional State Integration | Receive and process emotional embeddings to compute a composite Readiness Score. |
| FR3 | Adaptive Reminder Scheduling | Dynamically schedule reminders based on Readiness Score and item priority using RL agent. |
| FR4 | Content-Type Recommendation | Recommend appropriate study activity type based on current cognitive state and item difficulty. |
| FR5 | Reminder Delivery | Deliver reminders via mobile browser browser push notifications and in-page alert prompts with configurable timing. |
| FR6 | Feedback Collection | Collect student feedback (accept/snooze/dismiss) as RL reward signals for continuous learning. |
| FR7 | Student Profile Management | Maintain and update individual behavioral profiles, learning history, and scheduling preferences. |

*Table 5. Functional Requirements*

## 5.3 Non-Functional Requirements

A. Performance Requirements

- The system shall generate scheduling decisions within 500 milliseconds under normal load.
- The notification service shall deliver browser push notifications within 2 seconds of the scheduling decision.
- The system shall maintain consistent performance for at least 500 concurrent student users.

B. Accuracy and Reliability Requirements

- The adherence improvement shall exceed 85% compared to static scheduling baselines.
- The false positive reminder rate shall remain below 15%.
- The system shall maintain 99.5% uptime with graceful degradation if the emotion service is unavailable.

C. Usability Requirements

- The system shall achieve a System Usability Scale (SUS) score of at least 70.
- The reminder interface shall be non-intrusive and support simple one-tap feedback actions.

D. Security and Privacy Requirements

- Emotional embeddings shall not be stored in raw form; only aggregated Readiness Scores are persisted.
- All student data shall be encrypted at rest (AES-256) and in transit (TLS 1.3).
- The system shall comply with SLIIT's data protection and ethical research guidelines.

E. Scalability Requirements

- The architecture shall support horizontal scaling of scheduling and notification microservices.
- The system shall allow integration into larger EdTech platforms via documented RESTful APIs.

F. Ethical Requirements

- The system shall avoid generating biased or discriminatory scheduling decisions.
- The system shall maintain participant anonymity throughout data collection and analysis.
- The system shall comply with responsible AI guidelines and academic integrity standards.

# 6. COMMERCIALIZATION PLAN

## 6.1 Target Market and Customer Persona

Target Market

- Higher Education Institutions: Universities and institutes seeking AI-based academic support tools to improve student outcomes.
- EdTech Companies: Platforms integrating intelligent, personalized learning and wellness features.
- Online Learning Platforms: MOOCs and digital academies requiring affect-aware study management.
- Corporate Training Providers: Organizations offering structured technical upskilling programs.

Customer Persona

Name: Kavya – 21-year-old undergraduate student at SLIIT

- Goals: Prepare effectively for examinations, reduce study-related anxiety, avoid burnout.
- Pain Points: Existing apps send reminders at the wrong time (when exhausted or stressed), do not adapt to emotional state, and use the same reminder type regardless of mood.
- Your system directly addresses these issues by detecting emotional readiness and dynamically adjusting reminder timing, frequency, and content type.

## 6.2 Value Proposition

Core Value

A highly personalized study companion that respects the student's emotional state, avoids cognitive overload, and optimizes study timing and activity type for maximum retention with minimal fatigue.

Key Benefits

- Adherence: Reminders delivered at emotionally optimal moments improve accept rates by over 85%.
- Retention: Memory decay modeling ensures high-priority items are reviewed before the forgetting threshold.
- Wellbeing: Cognitive fatigue monitoring prevents burnout by reducing reminders during low-readiness periods.
- Personalization: Adaptive content-type selection matches study mode to emotional state.
- Integration: Seamlessly embedded into the Smart Study Assistant ecosystem and compatible with LMS platforms.

Why Institutions Will Adopt It

- Enhances student academic performance and reduces dropout rates.
- Provides actionable wellbeing analytics for academic advisors.
- Delivers a competitive advantage in student support services.

## 6.3 Revenue Model

1. Subscription Model (SaaS): Universities pay a monthly or annual fee based on enrolled student count, including updates, maintenance, and support.

2. Freemium Model: Basic reminder functionality free for students. Premium features (personalized analytics, cross-device sync) at LKR 500-800 per month.

3. Licensing Model: One-time license for institutions wanting on-premise deployment, with additional customization fees.

4. API-Based Usage Model: EdTech companies pay per API call or per 1,000 scheduling requests for integration into third-party platforms.

## 6.4 Cost Estimation

| Cost Category | Estimated Cost (LKR / month) |
|---|---|
| Cloud compute & storage | 40,000 – 70,000 |
| Development & maintenance | 30,000 – 50,000 |
| Licensing & tools | 0 – 10,000 (mostly open-source) |
| Support & monitoring | 10,000 – 15,000 |

*Table 6. Cost Estimation by Category*

## 6.5 Pricing Strategy

Tiered Pricing

- Basic Tier: Free – Limited reminders per day, single-device, no emotional analytics.
- Standard Tier: LKR 500/month – Full adaptive scheduling, multi-device, basic progress analytics.
- Premium Tier: LKR 800/month – Full analytics, institution-wide deployment, custom content integration.

Competitive Pricing

- Priced lower than global EdTech tools (Khanmigo, Chegg AI Tutor) to be affordable for Sri Lankan and regional markets.
- Institutional bulk pricing available for universities with more than 500 enrolled students.

## 6.6 Competitive Advantage

1. Emotion-Aware Scheduling: The first system to integrate affective readiness signals into adaptive scheduling, preventing reminders during high-stress or low-readiness periods.

2. Memory Decay Integration: The extended SM-2 algorithm with dynamic difficulty adjustment ensures scientifically optimal review timing at the item level.

3. Privacy-Preserving Design: Emotional embeddings processed on-device and never stored in raw form, addressing key adoption barriers in educational environments.

4. Reinforcement Learning Optimization: Continuous online learning from student feedback ensures the system improves personalization over time without manual retraining.

5. Local Market Fit: Designed specifically for Sri Lankan and regional education systems, making it more relevant and cost-effective than global tools.

## 6.7 Intellectual Property Considerations

The adaptive emotion-scheduling integration, RL-based reminder optimization pipeline, and the Readiness Score computation methodology represent novel contributions that can be protected under SLIIT's intellectual property framework. Course materials and student data remain the property of the university; only processed embeddings and aggregated profiles are retained by the system. The system complies with responsible AI guidelines and academic integrity standards.

# 7. BUDGET AND JUSTIFICATION

This section outlines the estimated costs required to design, develop, test, and evaluate the Emotion and Progress Aware Adaptive Study Reminder System. Costs are categorized into hardware, software, cloud services, and human effort.

## 7.1 Hardware Costs

| Item | Est. Cost (LKR) | Justification |
|---|---|---|
| Development Laptop / Workstation | 150,000 – 200,000 | Required for ML model experimentation, RL agent training, and backend development |
| External SSD Storage | 15,000 – 25,000 | Needed to store interaction logs, model checkpoints, and evaluation datasets |

*Table 7. Hardware Costs*

## 7.2 Software and Tool Costs

| Software / Tool | Est. Cost (LKR) | Justification |
|---|---|---|
| IDE Tools (VS Code, IntelliJ Community) | Free | Open-source tools sufficient for Python and Java development |
| Python ML Libraries (TF, PyTorch, scikit-learn) | Free | Open-source; industry-standard for ML and RL development |
| React.js SDK | Free | Open-source component-based web UI framework |
| API Testing Tools (Postman) | Free | Required for testing RESTful API endpoints |

*Table 8. Software and Tool Costs*

## 7.3 Cloud Services

| Service | Est. Cost (LKR/month) | Justification |
|---|---|---|
| Cloud Server (AWS EC2 / Firebase) – 6 months | 2,000/month | Required for hosting scheduling API and notification service |
| MongoDB Atlas – 6 months | 800/month | Managed database hosting for student profiles and interaction logs |
| Web Notification API (browser push) | Free tier | Reliable low-latency push notification delivery |

*Table 9. Cloud Service Costs*

## 7.4 Human Effort Cost

| Role | Est. Cost (LKR) | Justification |
|---|---|---|
| ML / RL Engineer | 50,000 – 80,000 | Designing memory decay model, RL agent, and emotion fusion module |
| Backend Developer | 30,000 – 50,000 | Implementing Spring Boot APIs, integration, and system architecture |
| Frontend Web Developer | 20,000 – 35,000 | Building React.js reminder delivery interface and feedback collection |
| QA / Tester | 15,000 – 25,000 | Testing scheduling accuracy, adherence metrics, and system reliability |

*Table 10. Human Effort Cost*

## 7.5 Budget Summary and Justification

| No. | Item Description | Unit Cost (LKR) | Total (LKR) |
|---|---|---|---|
| 1 | Cloud Server (AWS EC2 / Firebase) – 6 months | 2,000/month | 12,000 |
| 2 | MongoDB Atlas – 6 months | 800/month | 4,800 |
| 3 | Web Notification API (browser push) | Free | 0 |
| 4 | Python ML Libraries | Free | 0 |
| 5 | Data Collection Materials | 2,000 | 2,000 |
| 6 | User Study Participant Incentives (30 x LKR 200) | 200 each | 6,000 |
| 7 | Contingency Buffer (10%) | – | 2,480 |
| | **TOTAL** | | **27,280** |

*Table 11. Project Budget Summary*

Why the Budget Is Reasonable

- Most AI development and web development tools are open-source, significantly reducing licensing costs.
- Cloud usage is optimized through efficient microservice architecture and Firebase free tier for notifications.
- The contingency buffer ensures resilience against unexpected infrastructure or evaluation costs.

# 8. WORK BREAKDOWN STRUCTURE

The Work Breakdown Structure organizes all project activities into five phases: Research, Design, Development, Integration & Testing, and Evaluation & Report. Each phase is further decomposed into specific tasks aligned with the project objectives.

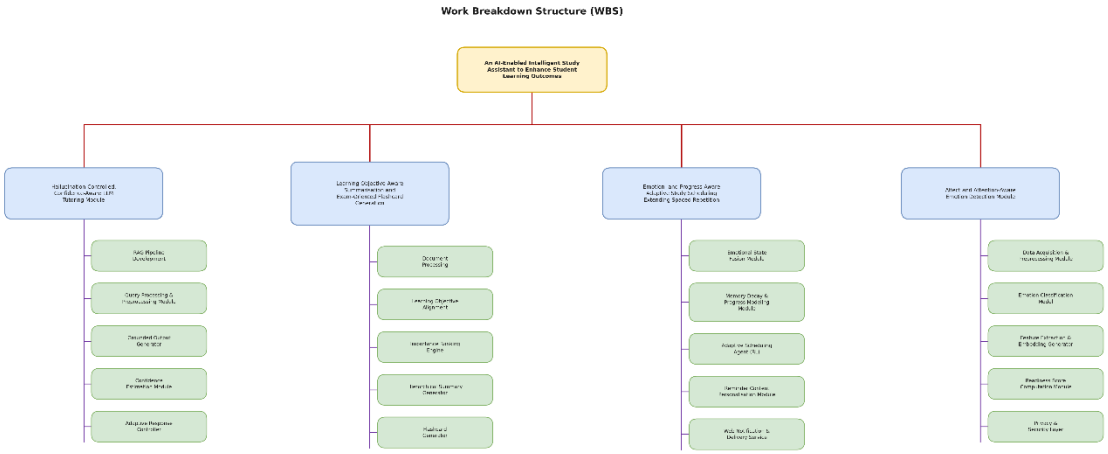



*Figure 3. Work Breakdown Structure – Adaptive Study Reminder System*

# 9. GANTT CHART

The project is scheduled for completion within 12 months (September 2025 – August 2026), with key milestones at PP1 (50% completion, Month 6) and PP2 (80% completion, Month 9).

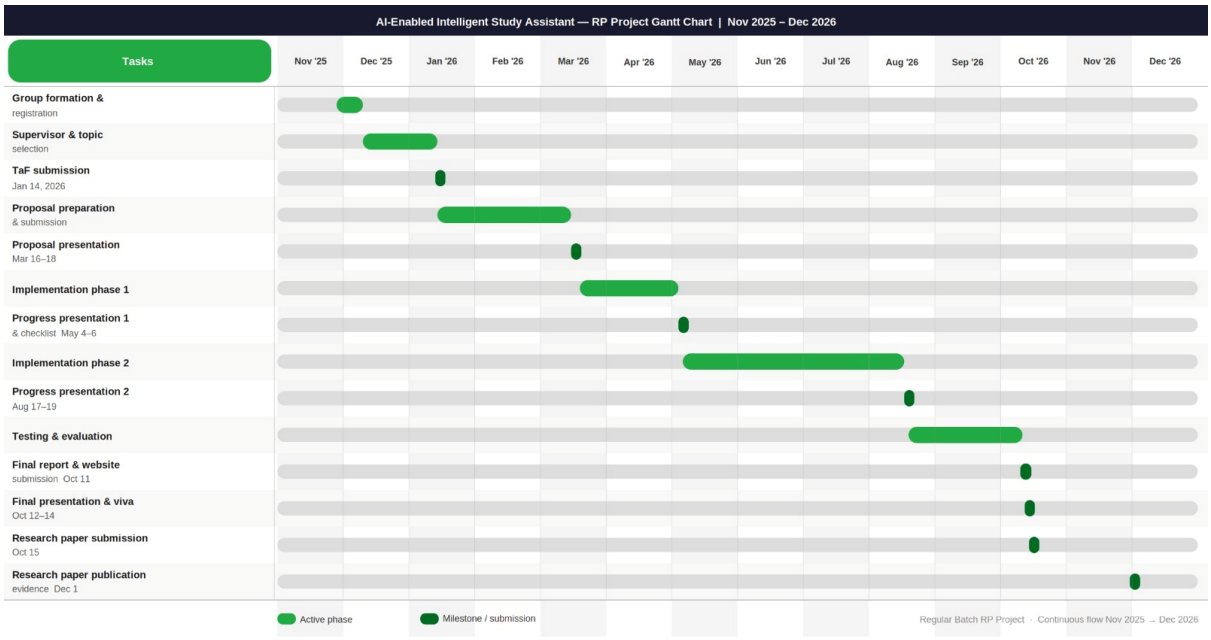



*Figure 4. Project Gantt Chart – September 2025 to August 2026*

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# APPENDICES

## Appendix A – Risk Analysis: Adaptive Study Reminder System

This appendix presents a systematic risk analysis for the Emotion and Progress Aware Adaptive Study Reminder System (R26-SE-022 – Component 4). Risks are identified across five categories: Technical, Integration, Research & Evaluation, Schedule, and Ethical & Privacy. Each risk is assigned a Likelihood (H = High, M = Medium, L = Low), an Impact rating (H/M/L), and a Severity level derived from their combination, together with a concrete mitigation strategy.

| ID | Risk Description | Category | Likelihood | Impact | Severity | Mitigation Strategy |
|---|---|---|---|---|---|---|
| **Technical Risks** | | | | | | |
| **R1** | RL scheduling agent fails to converge due to noisy or sparse emotional embedding inputs | Technical | M | H | **HIGH** | Implement epsilon-greedy exploration with a warm-up period using rule-based scheduling before RL takes over; add input validation and embedding normalisation at the fusion layer |
| **R2** | Extended SM-2 model produces inaccurate retention probability estimates for subject areas with irregular study patterns | Technical | M | M | **MEDIUM** | Validate SM-2 estimates against self-reported retention in the pilot study; dynamically tune difficulty parameters per student using online learning |
| **R3** | Web Push API not granted permission by the student's browser, blocking reminder delivery | Technical | M | M | **MEDIUM** | Provide an in-page alert fallback (React toast notifications) that does not require push permission; prompt users to enable notifications during onboarding |
| **R4** | React.js front-end performance degradation when rendering real-time updates for large student cohorts | Technical | L | M | **LOW** | Use React memoization (useMemo, React.memo) and lazy loading; implement server-sent events rather than polling for reminder triggers |
| **Integration Risks** | | | | | | |
| **R5** | Emotion Detection | Integration | M | H | **HIGH** | Define and agree a shared API contract (JSON schema) |

| ID | Risk Description | Category | Likelihood | Impact | Severity | Mitigation Strategy |
|---|---|---|---|---|---|---|
| | Module (Component 3) unavailable or delivers embeddings in an incompatible format | | | | | with Component 3 team by Phase 2; implement a graceful fallback that defaults to "Medium" readiness when emotion data is absent |
| **R6** | Shared Context Layer schema changes by another team member break the scheduling microservice's data contracts | Integration | L | M | **LOW** | Enforce versioned API contracts in the shared GitHub repository; use contract testing (Pact) during CI pipeline to catch breaking changes early |
| **Research & Evaluation Risks** | | | | | | |
| **R7** | Insufficient user study participation – fewer than 30 SLIIT students complete the 4-week evaluation | Research | M | H | **HIGH** | Recruit 40 participants (10 buffer) before the study begins; provide LKR 200 completion incentives; coordinate recruitment through SLIIT faculty with supervisor support |
| **R8** | Hawthorne effect – participants alter study behaviour because they know they are being observed, biasing adherence results | Research | M | M | **MEDIUM** | Use a between-subjects A/B design (control vs. emotion-aware group) and collect passive interaction logs to supplement self-report data; conduct a 1-week acclimatisation period before formal measurement begins |
| **R9** | Interaction log data is too sparse to train the RL agent meaningfully within the 4-week evaluation window | Research | L | M | **LOW** | Pre-train the contextual bandit on simulated interaction data generated from SM-2 patterns; switch to online learning once real data is available after week 1 |
| **Schedule Risks** | | | | | | |
| **R10** | RL agent development overruns its schedule, compressing the integration and evaluation phases | Schedule | M | M | **MEDIUM** | Begin RL development with a rule-based scheduling fallback as an MVP; use the Gantt milestones (PP1, PP2) as hard checkpoints to assess and re-plan if slippage is detected |

| ID | Risk Description | Category | Likelihood | Impact | Severity | Mitigation Strategy |
|---|---|---|---|---|---|---|
| **R11** | Delays in teammate module delivery (Components 1–3) block the integration phase and push back the user study | Schedule | M | M | **MEDIUM** | Develop and test this component independently using mock stubs for teammate APIs; integrate progressively once each module passes its unit tests |
| **Ethical & Privacy Risks** | | | | | | |
| **R12** | RL scheduling agent develops biased patterns that disadvantage students with consistently low emotional readiness scores | Ethical | L | H | **MEDIUM** | Implement a minimum reminder frequency floor so that low-readiness students still receive at least one study prompt per day; include fairness metrics in the evaluation dashboard |
| **R13** | Student interaction data or readiness scores are exposed due to a security breach in the cloud deployment | Privacy | L | H | **MEDIUM** | Enforce AES-256 encryption at rest, TLS 1.3 in transit, and JWT-based RBAC; store only aggregated Readiness Scores (never raw embeddings); pseudonymise all student IDs in logs |

### Severity Classification Key

| Severity | Likelihood + Impact | Action Required |
|---|---|---|
| **HIGH** | M+H or H+H | Immediate mitigation required |
| **MEDIUM** | M+M or L+H | Monitor and mitigate proactively |
| **LOW** | L+L or L+M | Accept and review at milestones |